

Chapter 3

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet for Chapter 3: Core AI Concepts for Beginners

This worksheet is designed for high school students (Class 9-12) as part of the course “AI & Future Work Preparedness.” It aligns with Chapter 3 content, covering key concepts such as Machine Learning, Data and Algorithms, AI Tools, Coding for AI, and Hands-On AI Model Building. The worksheet includes a variety of activities—practical tasks, creative exercises, and simple matching—to reinforce learning, encourage critical thinking, and make the material engaging. Students can complete this individually or in groups during class or as homework.

Part 1: Simple Matching Activity

Match the AI concept on the left with its correct description on the right by drawing a line between them.

AI Concept	Description
• 1. Machine Learning	• A. Step-by-step rules for processing data
• 2. Data	• B. AI tools like Siri or Alexa that help with tasks
• 3. Algorithms	• C. Teaching computers to learn from examples
• 4. Virtual Assistants	• D. Information or input that AI uses to learn

AI Concept

Description

- 1. Machine Learning
- A. Step-by-step rules for processing data
- 2. Data
- B. AI tools like Siri or Alexa that help with tasks
- 3. Algorithms
- C. Teaching computers to learn from examples
- 4. Virtual Assistants

- D. Information or input that AI uses to learn

Part 2: Practical Coding Task

Try this basic coding exercise using Python. You can use a free online tool like Google Colab (colab.research.google.com) or Replit. Write the code for the task below and note down what output you get. If you don't have access to a computer, write what you think the output would be.

Task: Write a small Python program to print a welcome message to AI learners. Use the `print()` function.

- Example Input: `print("Welcome to AI!")`
- Example Output: Welcome to AI!

Your Code:

Write your code here

`print(_____)` # Fill in the message

Output (or Predicted Output):

Reflection: What did you learn from writing this code? How could you change the message to something personal, like greeting a friend?

Part 3: Creative Thinking Exercise

Imagine you're tasked with designing a simple AI tool for your school. Answer the following to brainstorm your idea.

- **What problem will your AI tool solve?** (Example: Helping students track homework deadlines)
- **What kind of data would your AI need to work?** (Example: Student schedules and assignment lists)
- **How would students interact with this AI?** (Example: Through a chatbot on the school app)

Part 4: Hands-On Activity Reflection

If you completed the hands-on activity from the chapter (building a simple AI model to predict something like "pizza love"), answer these questions. If not, imagine you did and predict your experience.

- **What was the most interesting part of building or imagining a simple AI model?**
- **What challenges did you face (or think you would face), and how did/would you overcome them?**
- **How could you use a similar model for a real school project or personal idea? Give an example.**

Part 5: Fun Bonus Challenge (Optional)

AI Dream Project: If you could create any AI tool or app, what would it be? Draw a quick sketch or describe it below. Think about what it does, who uses it, and why it's cool. **Description or Sketch:**

(If sketching, use the space below or attach a separate sheet.)